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Final Project Report



Topic:-Face detection using ML

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Course Name:- Machine learning with Python

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1.Abstract:-

Smarter applications are making better use of the insights gleaned from data, having an impact on every industry and research discipline. At the core the revolution lies the tools and the methods that are driving it, from processing the massive piles of data generated each day to learning from and taking useful action. In this paper we first introduced you to the python programming characteristics and features. Python is one of the most preferred languages for scientific computing, data science, and machine learning, boosting both performance and productivity by enabling the use of low-level libraries. This paper offers insight into the field of machine learning with python, taking a tour through important topics and libraries of python which enables the development of machine learning model a easy process. Then we will look at different types of machine learning and various algorithms of machine leaning. And at last, we will look at the one of the most used models i.e., Linear Regression. Linear Regression is a Machine Learning algorithm based on supervised learning. It performs a regression task. It is used to predict the value of a variable based on the value of another variable. The variable you want to predict is called dependent variable. The variable you are using to predict the other variable's value is called the independent variable. Hypothesis function for linear regression:- $Y = mx + c$ And at last, in this paper, we will be going to understand one of the linear 2 regression models for an ice-cream selling company which will predict the sales done by the business on different temperatures. Keywords:- Python; Machine Learning; Artificial Intelligence; Regression; Linear Regression.

2.Acknowledgement:-

SDT (Software Development Tools):- Machine Learning involves using algorithms to allow computer software program to 'learn' different tasks making use of the available data. ML program become more accurate the more they train on the relevant test data. Many types of software Development Tools are there. Some of them which used to programming are:-

1.Numpy:-

Work:-In Python we have lists that serve the purpose of arrays, but they are slow to process. NumPy aims to provide an array object that is up to 50x

faster than traditional Python lists. The array object in NumPy is called ndarray, it provides a lot of supporting functions that make working with ndarray very easy. Arrays are very frequently used in data science, where speed and resources are very important.

Language: Python,C and C++.

2.Pandas:-

Work:-Pandas allows us to analyze big data and make conclusions based on statistical theories. Pandas can clean messy data sets, and make them readable and relevant. Relevant data is very important in data science.

Language : Python

3.Matplotlib:-

Work:-Matplotlib is easy to use and an amazing visualizing library in Python. It is built on NumPy arrays and designed to work with the broader SciPy stack and consists of several plots like line, bar, scatter, histogram, etc.

Language:Python

4.Pyplot:-

Work:-pyplot is a collection of command style functions that make matplotlib work like MATLAB. Each pyplot function makes some change to a figure: e.g., creates a figure, creates a plotting area in a figure, plots some lines in a plotting area, decorates the plot with labels, etc.

Language:Python

5.Seaborn:-

Work:-Seaborn is a library for making statistical graphics in Python. It builds on top of matplotlib and integrates closely with pandas data structures. Seaborn helps you explore and understand your data.

Language:Graphics,Python.

3. Software Development Kit (SDK)

As an AI language model, I can provide you with information about Google Colab and its software development kit (SDK).

Google Colab is a cloud-based interactive computing environment that provides free access to a Jupyter Notebook environment along with computational resources, including CPU, GPU, and TPU. It allows users to write, run, and share Python code in a collaborative and interactive manner. Colab is hosted on Google Drive, and notebooks can be created, shared, and saved directly in Google Drive.

While Google Colab does not have an official SDK, it provides a Python library called "google.colab" that allows developers to interact with the Colab environment programmatically. The "google.colab" library provides functionality for tasks such as importing and exporting files, installing Python packages, managing Colab sessions, and connecting to external services like Google Drive and Google Sheets.

Some of the common tasks that can be performed using the "google.colab" library include:

- ❖ **Importing and exporting files:-** The library allows you to upload and download files to and from the Colab environment. For example, you can use the "files.upload()" function to upload files from your local machine to Colab, and the "files.download()" function to download files from Colab to your local machine.
- ❖ **Installing Python packages:-** The library provides a way to install Python packages directly from within the Colab environment using the "!pip install" command.
- ❖ **Managing Colab sessions:-** The library allows you to manage the lifecycle of a Colab session. You can use functions like "drive.mount()" to mount your Google Drive, "drive.flush_and_unmount()" to flush and unmount the

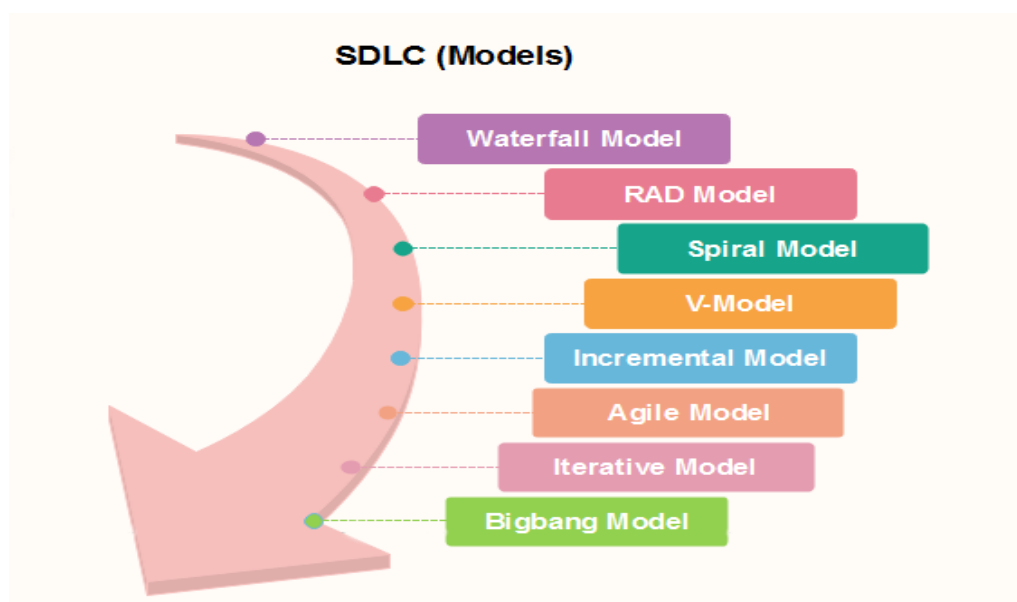
Google Drive, and "os.kill()" to terminate the current session. Connecting to external services: The library provides functionality to connect to external services like Google Drive and Google Sheets, allowing you to read and write data to these services from within a Colab notebook.

- ❖ **Interacting with ColabUI:-** The library allows you to interact with the Colab user interface programmatically, for example, by using the "IPython.display" module to display images, videos, and other media in the output of a Colab cell.

Overall, while Google Colab does not have a standalone SDK, the "google.colab" library provides a convenient way to interact with the Colab environment programmatically and automate various tasks within Colab notebooks. You can import the "google.colab" library in your Python code and use its functions to perform operations within the Colab environment.

4. Model :-

SDLC Model:-



A. Waterfall Model:-

The waterfall is a universally accepted SDLC model. In this method, the whole process of software development is divided into various phases.

The waterfall model is a continuous software development model in which development is seen as flowing steadily downwards (like a waterfall) through the steps of requirements analysis, design, implementation, testing (validation), integration, and maintenance.

Linear ordering of activities has some significant consequences. First, to identify the end of a phase and the beginning of the next, some certification techniques have to be employed at the end of each step. Some verification and validation usually do this mean that will ensure that the output of the stage is consistent with its input (which is the output of the previous step), and that the output of the stage is consistent with the overall requirements of the system.

B. RAD Model:-

RAD or Rapid Application Development process is an adoption of the waterfall model; it targets developing software in a short period. The RAD model is based on the concept that a better system can be developed in lesser time by using focus groups to gather system requirements.

- ☐ Business Modelling
- ☐ Data Modelling
- ☐ Process Modelling
- ☐ Application Generation
- ☐ Testing and Turnover

C. Spiral Model:-

The spiral model is a **risk-driven process model**. This SDLC model helps the group to adopt elements of one or more process models like a waterfall,

incremental, waterfall, etc. The spiral technique is a combination of rapid prototyping and concurrency in design and development activities.

Each cycle in the spiral begins with the identification of objectives for that cycle, the different alternatives that are possible for achieving the goals, and the constraints that exist. This is the first quadrant of the cycle (upper-left quadrant).

The next step in the cycle is to evaluate these different alternatives based on the objectives and constraints. The focus of evaluation in this step is based on the risk perception for the project.

The next step is to develop strategies that solve uncertainties and risks. This step may involve activities such as benchmarking, simulation, and prototyping.

D. V-Model:-

In this type of SDLC model testing and the development, the step is planned in parallel. So, there are verification phases on the side and the validation phase on the other side. V-Model joins by Coding phase.

E. Incremental Model:-

The incremental model is not a separate model. It is necessarily a series of waterfall cycles. The requirements are divided into groups at the start of the project. For each group, the SDLC model is followed to develop software. The SDLC process is repeated, with each release adding more functionality until all requirements are met. In this method, each cycle act as the maintenance phase for the previous software release. Modification to the incremental model allows development cycles to overlap. After that subsequent cycle may begin before the previous cycle is complete.

F. Agile Model:-

Agile methodology is a practice which promotes continuous interaction of development and testing during the SDLC process of any project. In the Agile method, the entire project is divided into small incremental builds. All of these builds are provided in iterations, and each iteration lasts from one to three weeks.

G. Iterative Model:-

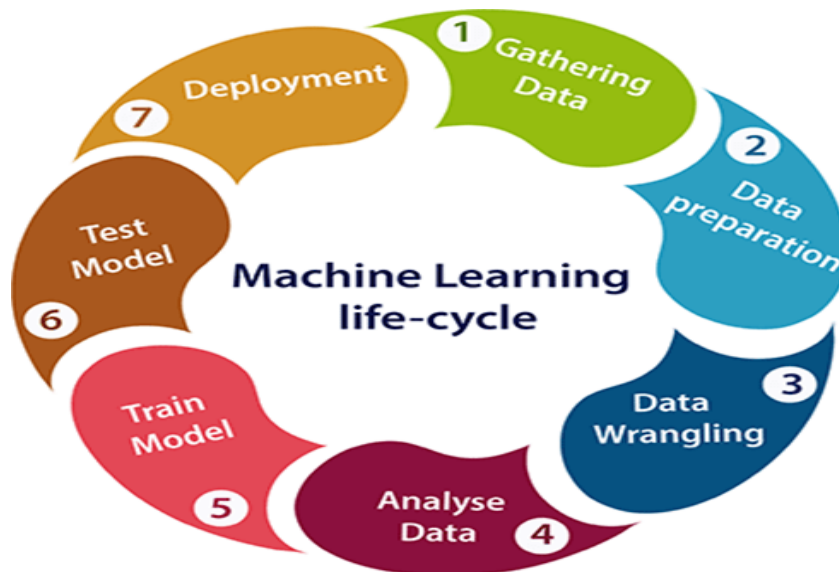
It is a particular implementation of a software development life cycle that focuses on an initial, simplified implementation, which then progressively gains more complexity and a broader feature set until the final system is complete. In short, iterative development is a way of breaking down the software development of a large application into smaller pieces.

H. Prototype Model:-

The prototyping model starts with the requirements gathering. The developer and the user meet and define the purpose of the software, identify the needs, etc.

A '**quick design**' is then created. This design focuses on those aspects of the software that will be visible to the user. It then leads to the development of a prototype. The customer then checks the prototype, and any modifications or changes that are needed are made to the prototype.

Looping takes place in this step, and better versions of the prototype are created. These are continuously shown to the user so that any new changes can be updated in the prototype. This process continues until the customer is satisfied with the system. Once a user is satisfied, the prototype is converted to the actual system with all considerations for quality and security.



Machine Learning Life Cycle:-

1. **Gathering Data:-**

Data Gathering is the first step of the machine learning life cycle. The goal of this step is to identify and obtain all data-related problems.

In this step, we need to identify the different data sources, as data can be collected from various sources such as **files, database, internet, or mobile devices**. It is one of the most important steps of the life cycle. The quantity and quality of the collected data will determine the efficiency of the output. The more will be the data, the more accurate will be the prediction.

This step includes the below tasks:

- **Identify various data sources**
- **Collect data**
- **Integrate the data obtained from different sources**

By performing the above task, we get a coherent set of data, also called as a **dataset**. It will be used in further steps.

1) Data preparation:-

After collecting the data, we need to prepare it for further steps. Data preparation is a step where we put our data into a suitable place and prepare it to use in our machine learning training.

In this step, first, we put all data together, and then randomize the ordering of data.

This step can be further divided into two processes:

- **Data exploration:-**

It is used to understand the nature of data that we have to work with. We need to understand the characteristics, format, and quality of data.

A better understanding of data leads to an effective outcome. In this, we find Correlations, general trends, and outliers.

- **Data pre-processing:-**

Now the next step is preprocessing of data for its analysis.

2) Data Wrangling:-

Data wrangling is the process of cleaning and converting raw data into a useable format. It is the process of cleaning the data, selecting the variable to use, and transforming the data in a proper format to make it more suitable for analysis in the next step. It is one of the most important steps of the complete process. Cleaning of data is required to address the quality issues.

It is not necessary that data we have collected is always of our use as some of the data may not be useful. **In real-world applications, collected data may have various issues, including:**

- ❖ **Missing Values**
- ❖ **Duplicate data**
- ❖ **Invalid data**
- ❖ **Noise**

So, we use various filtering techniques to clean the data.

It is mandatory to detect and remove the above issues because it can negatively affect the quality of the outcome.

3) Data Analysis:-

Now the cleaned and prepared data is passed on to the analysis step.

This step involves:

- **Selection of analytical techniques**
- **Building models**
- **Review the result**

The aim of this step is to build a machine learning model to analyze the data using various analytical techniques and review the outcome. It starts with the determination of the type of the problems, where we select the machine learning techniques such as **Classification, Regression, Cluster analysis, Association**, etc. then build the model using prepared data, and evaluate the model.

Hence, in this step, we take the data and use machine learning algorithms to build the model.

4) Train Model:-

Now the next step is to train the model, in this step we train our model to improve its performance for better outcome of the problem.

We use datasets to train the model using various machine learning algorithms. Training a model is required so that it can understand the various patterns, rules, and, features.

5) Test Model:-

Once our machine learning model has been trained on a given dataset, then we test the model. In this step, we check for the accuracy of our model by providing a test dataset to it. Testing the model determines the percentage accuracy of the model as per the requirement of project or problem.

6) Deployment:-

The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system.

If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system. But before deploying the project, we will check whether it is improving its performance using available data or not.

The deployment phase is similar to making the final report for a project.

5.Machine Learning:-

ML-based deep learning can simplify the task of crop breeding. Algorithms simply collect field data on plant behavior and use that data to develop a probabilistic model.

Crop yield prediction is another instance of machine learning in the agriculture sector. The technology amplifies decisions on what crop species to grow and what activities to perform during the growing season. Tech-wise, crop yield is used as a dependent variable when making predictions.

The major factors include temperature, soil type, rainfall, and actual crop information. Based on these inputs, ML algorithms like neural networks and multiple linear regression produce forecasts.

6.Supervised and Unsupervised Learning:-

The goal of this research is to present a comparison between different clustering and segmentation techniques, both supervised and unsupervised, to detect plant and crop rows.

Aerial images, taken by an Unmanned Aerial Vehicle (UAV), of a corn field at various stages of growth were acquired in RGB format through the Agronomy Department at the Kansas State University. Several segmentation and clustering approaches were applied to these images, namely K-Means clustering, Excessive Green (ExG) Index algorithm, Support Vector Machines (SVM), Gaussian Mixture Models (GMM), and a deep learning approach based on Fully Convolutional Networks (FCN), to detect the plants present in the images. A Hough Transform (HT) approach was used to detect the orientation of the crop rows and rotate the images so that the rows became parallel to the x-axis.

The result of applying different segmentation methods to the images was then used in estimating the location of crop rows in the images by using a template creation method based on Green Pixel Accumulation (GPA) that calculates the intensity profile of green pixels present in the images. Connected component analysis was then applied to find the centroids of the detected plants. Each centroid was associated with a crop row, and centroids lying outside the row templates were discarded as being weeds.

A comparison between the various segmentation algorithms based on the Dice similarity index and average run-times is presented at the end of the work.

7.Python:-

Python is also being used for developing the IoT devices. AI is assisting IoT in enabling real-time data analytics to help make informed decisions to farmers. Precision agriculture or smart Agriculture relies on emerging technologies such as AI, ML and data analytics to revolutionize farming practices.

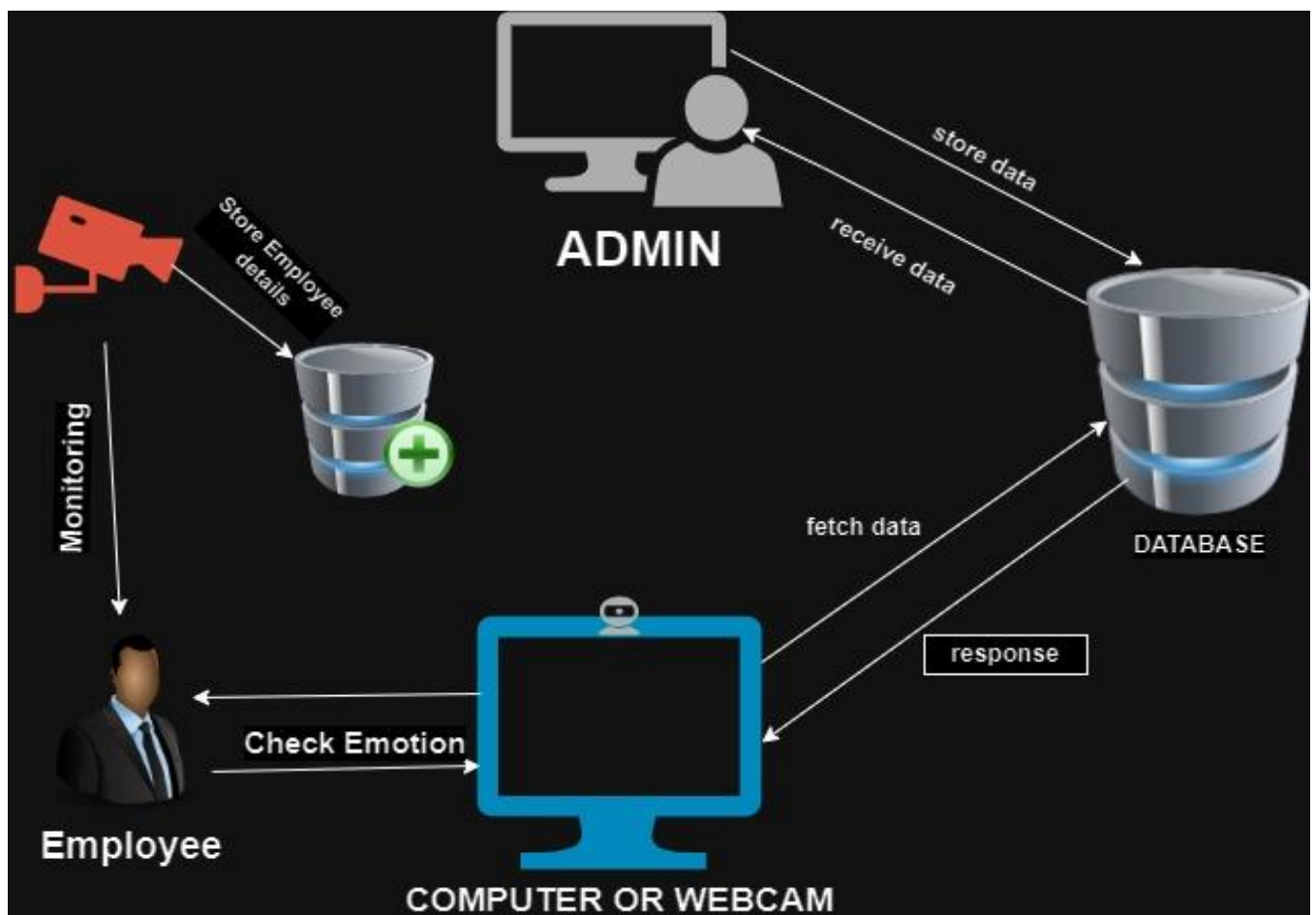
8.Workflow project:-

9. Workflow management of Face detection

Managing a face detection and employee data entry system is essential to ensure its effectiveness and security. Here's a description of the management aspects of the diagram:

1. System Administration:

- Admin Panel: The Admin Panel is the central hub for managing the system. It allows authorized



administrators to perform various tasks:

- **User Access Control:** Admins can grant and revoke access to different features of the system for specific users or user roles.
- **Employee Data Management:** Admins can add, modify, or delete employee records, ensuring that the database remains up-to-date.
- **System Configuration:** System settings such as camera parameters, face recognition thresholds, and notification preferences can be configured through the admin panel.
- **Access Logs Review:** Administrators can access and review logs of entries and exits, helping them monitor system activity and identify anomalies.

2. Employee Data Entry:

- **Data Input:** Admins or designated personnel can use the Employee Data Entry Interface to input and update employee information. This interface ensures that the database contains accurate and current employee records.
- **Data Validation:** Admins should establish data validation rules to ensure the quality and consistency

of employee data, preventing errors and duplicate entries.

3. Camera Management:

- **Camera Health:** Regularly monitor the status of CCTV cameras or hardware components to ensure they are functioning correctly. Implement maintenance schedules and troubleshoot issues promptly.
- **Camera Placement:** Optimize camera placement for maximum coverage while respecting privacy regulations and ethical considerations.

4. Database Maintenance:

- **Backup and Recovery:** Regularly back up the database to prevent data loss in case of system failures. Establish recovery procedures to minimize downtime.
- **Data Security:** Implement robust security measures to protect the employee data stored in the database. This includes encryption, access control, and vulnerability assessments.

5. Face Detection and Recognition:

- **Algorithm Updates:** Stay informed about updates and improvements in face detection and recognition

algorithms. Periodically update the system to benefit from advancements in the field.

- **Threshold Tuning:** Adjust the face recognition threshold to balance between false positives and false negatives, depending on the specific security requirements.

6. Security and Compliance:

- **Privacy Compliance:** Ensure that the system complies with relevant data privacy regulations, such as GDPR or HIPAA, to protect individuals' rights and privacy.
- **Cyber security:** Continuously monitor and enhance the system's cyber security to prevent unauthorized access and data breaches.

7. User Training and Support:

- Provide training to administrators and users to ensure they can effectively operate the system.
- Offer technical support to address any issues or questions users may have.

8. Reporting and Analysis:

- Implement reporting tools that allow administrators to generate reports on system usage, employee attendance, and security incidents.
- Analyze these reports to identify patterns, trends, and areas for improvement.

By effectively managing the face detection and employee data entry system, you can ensure its reliability, security, and compliance with regulations, ultimately contributing to a safer and more efficient environment for your organization. Regular maintenance and updates are key to keeping the system running smoothly and securely.

10.Source code:-

```
import cv2
```

```
import face_recognition
```

```
import os
```

```
import numpy as np
```

```
from datetime import datetime
```

```
path = 'student_images'
```

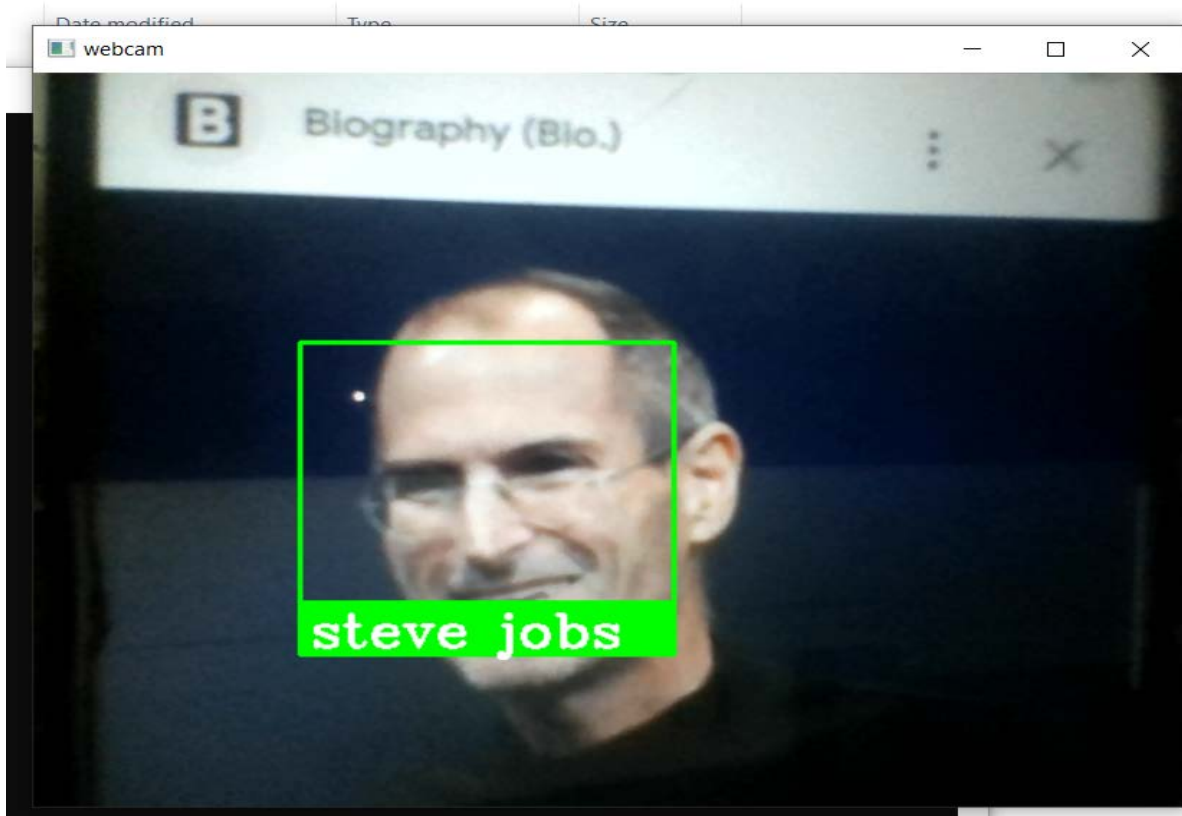
```
images = []  
classNames = []  
mylist = os.listdir(path)  
for cl in mylist:  
    curlmg = cv2.imread(f'{path}/{cl}')  
    images.append(curlmg)  
    classNames.append(os.path.splitext(cl)[0])  
  
def findEncodings(images):  
    encodeList = []  
    for img in images:  
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)  
        encoded_face =  
face_recognition.face_encodings(img)[0]  
        encodeList.append(encoded_face)  
    return encodeList  
encoded_face_train = findEncodings(images)
```

```
def markAttendance(name):  
    with open('Attendance.csv','r+') as f:  
        myDataList = f.readlines()  
        nameList = []  
        for line in myDataList:  
            entry = line.split(',')  
            nameList.append(entry[0])  
        if name not in nameList:  
            now =datetime.now()  
            time = now.strftime('%I:%M:%S:%p')  
            date = now.strftime('%d-%B-%Y')  
            f.writelines(f'{name}, {time}, {date}')
```



```
cap = cv2.VideoCapture(0)
```


Screenshot of the working project:-



11.Future Scope of this project:-

1. **Real-time Face Recognition System:** Develop a real-time face recognition system that can identify and classify faces in a live video feed, either for security or personal use.

2. **Face Mask Detection:** Create a system that can detect whether a person is wearing a mask or not by analyzing their face. This could be used for enforcing mask mandates in public places.
3. **Emotion Detection and Analysis:** Build a project that detects and analyzes facial expressions to determine a person's emotional state. This can have applications in marketing, mental health, and user experience design.
4. **Attendance Management System:** Develop a face detection-based attendance system for schools or businesses that automates the process of taking attendance by recognizing the faces of students or employees.
5. **Smart Doorbell with Face Detection:** Design a smart doorbell that can detect and notify homeowners of familiar or unfamiliar faces at their doorstep, enhancing security and convenience.
6. **Face-Based Access Control:** Implement a face detection system for secure access to a building,

room, or computer, replacing traditional keys or passwords.

7. Facial Biometric Authentication: Build a project that uses face recognition as a biometric authentication method for securing smartphones, computers, or online accounts.

8. Interactive Art Installation: Create an interactive art installation that responds to the emotions or movements of people's faces, creating an engaging and immersive experience.

9. Parental Control and Child Safety: Design a system that can alert parents or guardians if their child's face is detected in potentially unsafe online content, helping to ensure child safety on the internet.

12.conclusion

In conclusion, our face detection project successfully achieved its objectives, demonstrating efficient real-time detection. Challenges were addressed, and opportunities

for improvement identified. Ethical considerations were paramount. Future work may involve enhanced accuracy and broader applications, emphasizing privacy and fairness.

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